

SERVICE COSTING

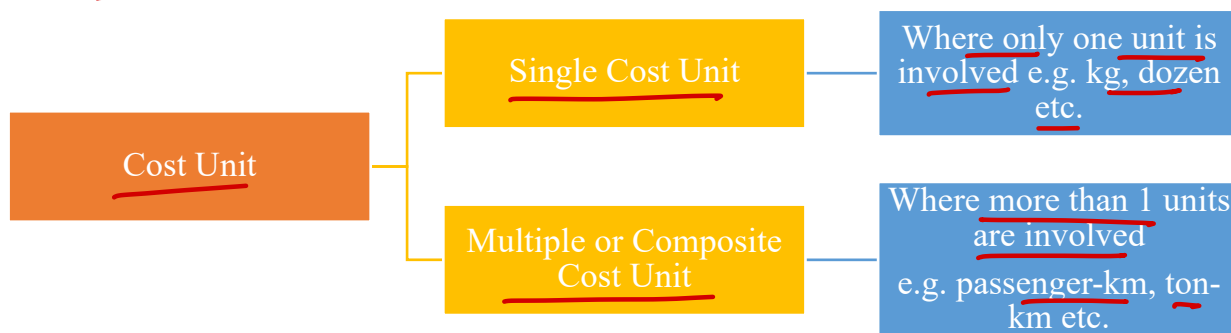
1. Service Costing

It can be internally or externally.

Industries will have high fixed cost than variable cost.

2. Cost Unit

The terms in which costs are expressed.



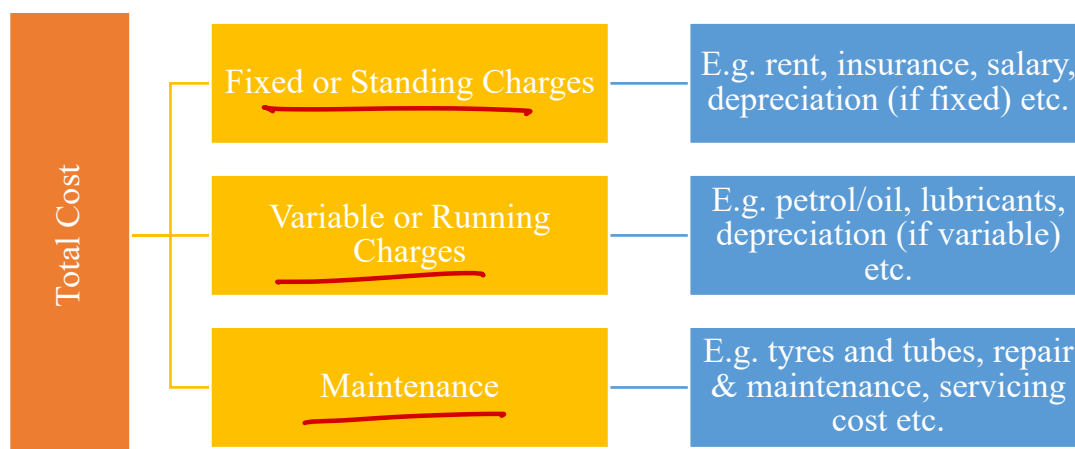
3. Cost Unit

(A) **Absolute Ton-km** = Actual kms × Actual tons

(B) **Commercial Ton-km** = Actual kms × Average tons

4. **Effective Cost Unit** = Total units – Normal loss units

5. Total Cost

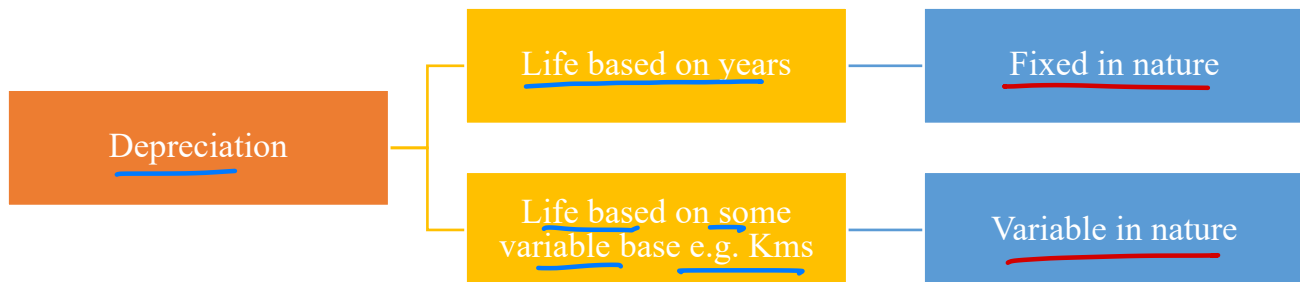


6. Cost Per Unit = $\frac{\text{Total cost}}{\text{Effective cost unit}}$ ✓

7. **Points To Remember (PTRs)**

(A) Petrol, oil and similar charges are always on the basis of actual km travel

(B) Treatment of Depreciation



(C) **Calculation of Service Cost**

Service cost of Rs. 4,000 after every 3000 kms.

If actual kms travel; = 3,500;

No. of service = $3500/3000 = 1.17$ or 1

Cost = $4000 \times 1 = 4,000$

If actual kms travel; = 8,500;

No. of service = $8500/3000 = 2.83$ or 2

Cost = $4000 \times 2 = 8,000$

(D) In case of different charges for different categories of service, use the concept of equivalent unit of services.

	<u>Ratio</u>	<u>No. of seats</u>	<u>Equiv. No. Ticket</u>
Normal	1	100	100
Premium	2	150	300
Recline	4	20	80
			<u>480</u>

$$\begin{array}{rcl}
 \text{TC of a Show} & = & \text{Rs. } 5000 \\
 (+) \text{ Pft. @ } 20\% & = & \text{Rs. } 1000 \\
 \hline
 \text{Sale} & = & \text{Rs. } 6000
 \end{array}$$

No. ticket Rate = $6000 \div 480 = 12.50$

Prem. " " = $12.50 \times 2 = 25$

Recline " " = $12.50 \times 4 = 50$

Question – 1

Mr. S owns a bus which runs according to the following schedule:

- (a) Delhi to Panchkula and back, the same day

Distance covered: 150 kms one way

Number of days run each month: 8

Seating capacity occupied 90%

- (b) Delhi to Mathura and back, the same day

Distance covered: 120 kms one way

Number of days run each month: 10

Seating capacity occupied 85%

- (c) Delhi to Alwar and back, the same day

Distance covered: 270 kms one way

Number of days run each month: 6

Seating capacity occupied 100%

- (d) Following are the other details:

Cost of the bus → Dep. - F

Salary of the driver → F

Salary of the conductor → F

Salary of the part-time accountant → F

Insurance of the bus → F

Diesel consumption 4 kms per litre → V

Road tax → F

Lubricant oil → V

Permit fee → F

Repairs and maintenance → M

Depreciation of the bus → F

Seating capacity of the bus

No.	x Kms	x Pass.	= Pass-km
1	x 150 x 2 x 8	x 50 x 90% =	
1	x 120 x 2 x 10	x 50 x 85% =	
1	x 270 x 2 x 6	x 50 x 100% =	
Kms = 8040			1 month

$$\begin{aligned}
 \text{TC} &= 29904 \\
 (+) \text{ Profit } [30\% \times 59808] &= \\
 (+) \text{ Pass-tax } [20\% \times 59808] &= \\
 \hline
 \text{Totalings } [29904 \div 50\%] &= 59808 \\
 \text{Pass-km} &= \\
 \hline
 \text{Totalings Pass-km} &=
 \end{aligned}$$

₹ 6,00,000

₹ 2,800 p.m.

₹ 2,200 p.m.

₹ 200 p.m.

₹ 4,800 p.a. ÷ 12 = 400

₹ 6 per litre → $\frac{6}{4} \times 8040 = 12060$

₹ 1,500 p.a. ÷ 12 = 125

₹ 10 per 100 kms → $\frac{10}{100} \times 8040 = 804$

₹ 315 p.m.

₹ 1,000 p.m.

@ 20% p.a. → $62 \times 20\% \times \frac{1}{12} = 10000$

→ 50 persons

Passenger tax is 20% of the total takings. Calculate the bus fare to be charges from each passenger to earn a profit of 30% on total takings. The fares are to be indicated per passenger for the journeys:

- (i) Delhi to Panchkula; (ii) Delhi to Mathura; and (iii) Delhi to Alwar

Solution

Calculation of Passenger Kms

	No.	x	Kms	x	Passenger	=	Passenger Kms
Delhi – Panchkula	1	x	150 x 2 x 8	x	50 x 90% ✓	=	1,08,000
Delhi – Mathura	1	x	120 x 2 x 10	x	50 x 85% ✓	=	1,02,000
Delhi – Alwar	1	x	270 x 2 x 6	x	50 x 100% ✓	=	1,62,000
Total							3,72,000

Kms travel = $(1 \times 150 \times 2 \times 8) + (1 \times 120 \times 2 \times 10) + (1 \times 270 \times 2 \times 6) = 8,040 \text{ kms}$

Statement of Operating Cost

Particulars	Amount (₹)
Fixed Cost:	
Depreciation $(6,00,000 \times 20\% \times 1/12)$	10,000
Driver salary	2,800
Conductor Salary	2,200
Accountant salary	200
Insurance $(4,800 \times 1/12)$	400
Road Tax $(1,500 \times 1/12)$	125
Permit Fee	315
Total Fixed Cost (A) →	16,040
Variable Cost:	
Diesel $[8,040 \times (6/4)]$	12,060
Lubricating oil $[8,040 \times (10/100)]$	804
Total Variable Cost (B) →	12,864
Maintenance Cost:	
Repair & Maintenance	1,000
Total Maintenance Cost (B)	1,000
Total Cost (A + B + C)	29,904
(+) Passenger tax $(59,808 \times 20\%)$	11,962
(+) Profit $(59,808 \times 30\%)$	17,942
Total Takings $(29,904 \div 50\%)$	59,808
Effective Passenger km →	3,72,000
Takings per effective passenger km	0.16

Fares per passenger for the journey

Delhi to Panchkula = $150 \times 0.16 = ₹ 24$ Delhi to Mathura = $120 \times 0.16 = ₹ 19.20$ Delhi to Alwar = $270 \times 0.16 = ₹ 43.20$

8 km → } Sr. } Morning 8
 ← } Jr. }
 ← }
 ← } Jr. } Afternoon
 ← } Sr. }

Question – 2

SK is a public school having five buses each plying on different directions for the transport of its school students. In view of a large number of students available of the bus service the buses work two shifts daily both in the morning and in afternoon. The buses are garaged in the school. The work load of the students has been so arranged that in the morning the first trip picks up the senior students and the second trip plying an hour later picks up the junior students. Similarly, in the afternoon the first trip picks the junior students and an hour later the second trip takes the senior student home.

The distance travelled by each bus one way is 8 kms. The school works 25 days in a month and remains closed for the vacation in May, June and December. Bus fee however is payable by the students for all the 12 months of the year.

The details of expenses for a year are as under:

Dist. travel Per day = $8 \times 8 = 64 \text{ km}$
 Km p.a. = $64 \times 25 \times 9 = 14400 \text{ km}$

Fee Per Stud. = $50 + 50 = 100$

Half fare stud. = 50
 Full " " = 50
 Sq. Half fare stud. = $50 + (50 \times 2) = 150$

Cost Per Half fee Std. = $\frac{2400}{150} = 16$
 Full fee = $16 \times 2 = 32$

TC Per bus p.a. = 28800 ✓ TC P.M. = $\frac{28800}{12} = 2400$

Drivers' salary (F)	₹ 450 per month per driver $\times 12 = 5400$
Cleaners' salary (F)	₹ 350 per month $\times 12 \times \frac{1}{2} = 840$
(Salary payable for 12 months and one cleaner employed for all the five buses)	
License fee, taxes etc. (F)	₹ 860 per bus per annum ✓
Insurance (F)	₹ 1,000 per bus p.a. ✓
Repairs and maintenance (M)	₹ 3,500 per bus p.a. ✓
Purchase price of bus (life 12 years) Dep. (F)	₹ 1,50,000 each $\rightarrow \frac{1,50,000 - 30,000}{12} = 10,000$
Scrap value	₹ 30,000
Diesel cost (✓)	₹ 2.00 per litre $\rightarrow \frac{2}{4} \times 14,400 = 7200$

Each bus gives an average mileage of 4 km per liter of diesel.

Seating capacity of each bus is 50 students. The seating capacity is fully occupied during the whole year. Students picked up and dropped within a range upto 4 km of distance from the school are charged half fare and fifty percent of students travelling in each trip are in this category. Ignore interest since the charges are to be based on average cost.

You are required to:

- Prepare a statement showing the expenses of operating a single bus and the fleet of five buses for a year. 28800 $1,44,000$
- Work out the average cost per student per month in respect of
 - Students coming from a distance of upto 4 km from the school and
 - Students coming from a distance of beyond 4 km from the school

Solution

(a) Statement of Cost

Particulars	1 bus	5 buses
Driver salary	$450 \times 12 = 5,400$	$5,400 \times 5 = 27,000$
Cleaner salary	$4,200 \div 5 = 840$	$350 \times 12 = 4,200$
License etc.	860	$860 \times 5 = 4,300$
Insurance	1,000	$1,000 \times 5 = 5,000$
Repair & Maintenance	3,500	$3,500 \times 5 = 17,500$
Depreciation	$(1,50,000 - 30,000) \div 12 = 10,000$	$10,000 \times 5 = 50,000$
Diesel (w.n. - 1)	7,200	$7,200 \times 5 = 36,000$
Total	28,800	1,44,000

Working note 1

Number of round trips in a day = 8

Total kms in a day = $8 \times 8 = 64$

Total kms per annum = $64 \times 25 \times 9 = 14,400$

Diesel cost per annum per bus = $14,400 \times (2/4) = ₹ 7,200$

(b) Number of students from whom bus fee is receivable = 100

Half fee students per bus = $100 \times 50\% = 50$

Full fee students per bus = $100 \times 50\% = 50$

Full fee students equivalent to half fee students = $50 \times 2 = 100$

Total equivalent half fee students = $50 + 100 = 150$

Total cost per bus per annum = ₹ 28,800

Total cost per bus per month = $28,800 \div 12 = ₹ 2,400$

Total cost per month per half fee students = $2,400 \div 150 = ₹ 16$

Total cost per month per full fee students = $16 \times 2 = ₹ 32$

Question – 3

A company is considering three alternative proposals for conveyance facilities for its sales personnel who have to do considerable travelling, approximately 20,000 kms every year. The proposal are as follows:

- Purchase and maintain its own fleet of cars. The average cost of a car is ₹ 1,00,000
- Allow the executive use his own car and reimburse expenses at the rate of ₹ 1.60 paise per kilometer and also bear insurance costs.
- Hire cars from an agency at ₹ 20,000 per year per car. The company will have to bear costs of petrol, taxes and tyres.

The following further details are available:

✓ Petrol ₹ 0.60 per km

✓ Repairs and maintenance ₹ 0.20 per km

✓ Tyre ₹ 0.12 per km

✗ Insurance ₹ 1,200 per car per annum;

✗ Taxes ₹ 800 per car per annum

✗ Life of the car: 5 years with annual mileage of 20,000 kms

✗ Resale value: ₹ 20,000 at the end of the fifth year

Work out the relative costs of three proposals and rank them.

	Own	Ex.	Hire
Cost of car	1,00,000	1,00,000	-
Depreciation	16,000	-	-
Insurance	1,200	1,200	-
Taxes	800	-	800
Hire charges	-	-	20,000
Total Fixed Cost	18,000	1,200	20,800
Kms travel	20,000	20,000	20,000
Fixed cost per km (A)	0.90	0.06	1.04

Solution

Particulars	Own car	Executive car	Hire car
Fixed Cost:			
Depreciation $[(1,00,000 - 20,000) \div 5]$	16,000	-	-
Insurance	1,200	1,200	-
Taxes	800	-	800
Hire charges	-	-	20,000
Total Fixed cost	18,000	1,200	20,800
Kms travel	20,000	20,000	20,000
Fixed cost per km (A)	0.90	0.06	1.04

Particulars	Own car	Executive car	Hire car
Variable Cost per km:			
Petrol	0.60	-	0.60
Repair & Maint.	0.20	-	-
Tyres	0.12	-	0.12
Reimbursement	-	1.60	-
Total variable cost per km (B)			
Total cost per km (A + B)	1.82	1.66	1.76
Rank	III	I	II

Question – 4

A LMV Pvt. Ltd., operates cab/car rental service in Delhi/NCR. It provides its service to the offices of Noida, Gurugram and Faridabad. At present it operates CNG fueled cars but it is also considering to upgrade these into Electric vehicles (EV). The details related with the owning of CNG & EV propelled cars are as tabulated below:

Particulars	CNG Car	EV Car
Car purchase price (₹)	9,20,000	15,20,000
Govt. subsidy on purchase of car (₹)	$\frac{9.20 - 0.95}{15} = 55000$	$\frac{15.20 - 1.50 - 1.70}{10} = 1.20$
Life of the car	15 years	10 years
Residual value (₹)	95,000	1,70,000
Mileage	20 km/kg	240 km per charge
Electricity consumption per full charged	$\frac{60}{20} \times$	30 Kwh
CNG cost per Kg (₹)	60	$\frac{30 \times 7.6}{240} \times$
Power cost per Kwh (₹)	-	7.60
Annual Maintenance cost (₹)	8,000	5,200
Annual insurance cost (₹)	7,600	14,600
Tyre replacement cost in every 5 year (₹)	$\frac{16000 \times 2}{15} = 2133$	$\frac{16000 \times 1}{10} = 1600$
Battery replacement cost in every 8 years (₹)	$\frac{12000 \times 1}{15} =$	$\frac{5.40 \times 1}{10} =$

Apart from the above, the following are the additional information:

Particulars	
Average distance covered by a car in a month	→ 1,500 km
Driver's salary (₹)	→ 20,000 p.m.
Garage rent per car (₹)	→ 4,500 p.m.
Share of office and administration cost per car (₹)	→ 1,500 p.m.

You have been approached by the management of A LMV Pvt. Ltd. for consultation on the two options of operating the cab service. The expected questions that may be asked by the management are as follows:

Question - 1

What would be the depreciable value of CNG car and EV car respectively?

- (a) ₹ 13,50,000 and ₹ 14,40,000
 (b) ₹ 15,20,000 and ₹ 8,25,000
 (c) ₹ 8,25,000 and ₹ 14,40,000
 (d) ₹ 8,25,000 and ₹ 12,00,000

$$9.208 - 0.950 = 8.258$$

$$15.208 - 1.508 - 1.700 = 120$$

Question - 2

What would be the monthly cost of fuel and electricity for an CNG and EV care respectively?

- (a) ₹ 4,500 and ₹ 1,425
 (b) ₹ 1,500 and ₹ 4,500
 (c) ₹ 1,525 and ₹ 1,450
 (d) ₹ 1,525 and ₹ 1,425

$$\frac{60}{20} \times 1500 = 4500$$

$$\frac{30 \times 7.6}{240} \times 1500 = 1425$$

Question - 3

What would be the total cost to be incurred for replacement of tyres for CNG and EV care respectively?

- (a) ₹ 32,000 and ₹ 24,000
 (b) ₹ 12,000 and ₹ 32,000
 (c) ₹ 32,000 and ₹ 16,000
 (d) ₹ 16,000 and ₹ 12,000

$$16000 \times 2 = 32000$$

$$16000 \times 1 = 16000$$

Question - 4

What would be the total cost to be incurred for replacement of battery for CNG and EV car respectively?

- (a) ₹ 5,40,000 and ₹ 12,000
 (b) ₹ 12,000 and ₹ 5,40,000
 (c) ₹ 2,00,000 and ₹ 12,000
 (d) ₹ 1,00,000 and ₹ 2,00,000

$$12000 \times 1 = 12000$$

$$5.400 \times 1 = 5.400$$

Question - 5

What would be the operating cost of vehicle per month per car for both CNG and EV options?

- (a) ₹ 36,627.78 and ₹ 43,708.33
 (b) ₹ 36,627.78 and ₹ 48,523.26
 (c) ₹ 48,523.26 and ₹ 28,150.29
 (d) ₹ 48,523.26 and ₹ 28,510.29

$$36627.75$$

$$43708.33$$

1	2	3	4	5
D	A	C	B	A

$$\begin{array}{l} \text{Pilot} \rightarrow \times \\ \text{Plot} \leftarrow \times \end{array} \begin{array}{l} \text{No.} \\ 1 \\ 1 \end{array} \times \begin{array}{l} \text{Km} \\ 15 \\ 15 \end{array} \times \begin{array}{l} \text{Ton} \\ 0 \\ 4 \end{array} = \begin{array}{l} \text{Ton-km} \\ 0 \\ 60 \end{array}$$

$$\begin{array}{l} \text{Plot} \rightarrow \times \\ \text{Plot} \leftarrow \times \end{array} \begin{array}{l} 1 \\ 1 \end{array} \times \begin{array}{l} 20 \\ 20 \end{array} \times \begin{array}{l} 0 \\ 4 \end{array} = \begin{array}{l} 0 \\ 80 \end{array}$$

$$\begin{array}{l} \text{Plot} \rightarrow \times \\ \text{Plot} \leftarrow \times \end{array} \begin{array}{l} 1 \\ 1 \end{array} \times \begin{array}{l} 20 \\ 20 \end{array} \times \begin{array}{l} 0 \\ 4 \end{array} = \begin{array}{l} 0 \\ 80 \end{array}$$

$$\begin{array}{l} \text{Travel} \\ \text{Load} \\ \text{Unload} \\ \text{Time} \end{array} \begin{array}{l} \frac{60}{40} \times 30 = 45 \text{ min.} \\ 30 \text{ min.} \\ 15 \text{ min.} \\ 90 \text{ min.} \end{array} \begin{array}{l} \frac{4}{40} \times 40 = 60 \text{ min.} \\ 25 \text{ min.} \\ 15 \text{ min.} \\ 100 \text{ min.} \end{array}$$

Question – 5

Coal is transported from two mines X & Y and unloaded at plots in a railway station. X is at distance of 15 kms and Y is at a distance of 20 kms from the rail head plots. A fleet of lorries having carrying capacity of 4 tonnes is used to transport coal from the mines. Records reveal that average speed of the lorries is 40 kms per hour when running and regularly take 15 minutes to unload at the rail head. At Mine X average loading time is 30 minutes per load, while at mine Y average loading time is 25 minutes per load.

Additional information:

Drivers' wages, depreciation, insurance and taxes, etc. ₹ 12 per hour

Operated Fuel, oil tyres, repairs and maintenance, etc. ₹ 1.60 per km

$$\begin{array}{l}
 \text{F} \quad \frac{12}{60} \times 90 = 18 \quad \frac{12}{60} \times 100 = 20 \\
 \text{V} \quad 1.60 \times 30 = 48 \quad 1.60 \times 40 = 64 \\
 \text{TC} \quad 66 \quad 84 \\
 \text{Ton km} \quad 60 \quad 80 \\
 \text{TC/Ton km} \quad 1.10 \quad 1.05
 \end{array}$$

You are required to prepare a statement showing the cost per tonne kilometer of carrying coal from each mine 'X and 'Y'.

Solution**Calculation of Ton Kms**

	No.	×	Kms	×	Ton	=	Ton Kms
Plot to X	1	×	15	×	0	=	0
X to plot	1	×	15	×	4	=	60
Total						=	60

$$\text{Kms travel} = (1 \times 15) + (1 \times 15) = 30$$

Calculation of Ton Kms

	No.	×	Kms	×	Ton	=	Ton Kms
Plot to Y	1	×	20	×	0	=	0
Y to Plot	1	×	20	×	4	=	80
Total						=	80

$$\text{Kms travel} = (1 \times 20) + (1 \times 20) = 40$$

Statement to time (in minutes)

Particulars	X	Y
Travel time	$30 \times (60/40) = 45$	$40 \times (60/40) = 60$
Loading time	30	25
Unloading time	15	15
Total time	90	100

Statement of operating cost

Particulars	X	Y
Fixed cost	$90 \times (12/60) = 18$	$100 \times (12/60) = 20$
Variable cost	$1.60 \times 30 = 48$	$1.60 \times 40 = 64$

Total Cost	66	84
Ton-Kms	60	80
Total cost per ton-km	1.10	1.05

Question – 6

SK Transport Ltd. charges ₹ 90 per ton for its 6 tons truck lorry load from city 'A' to city 'B'. The charges for the return journey are ₹ 84 per ton. No concession or reduction in these rates is made for any delivery of goods at intermediate station 'C'. In January, the truck made 12 outward journeys for city 'B' with full load out of which 2 ton were unloaded twice in the way at city 'C'. The truck carried a load of 8 tons in its return journey for 5 times but once caught by police and ₹ 1,200 was paid as fine. For the remaining trips the truck carried full load out of which all the goods on load were unloaded once at city 'C'.

The distance from city 'A' to city 'C' and city 'B' are 140 kms and 300 kms respectively. Annual fixed cost and maintenance charges are ₹ 60,000 and ₹ 12,000 respectively. Running charges spent during January, are ₹ 2,944. You are required to find out the cost per absolute ton-km and the profit for January.

Solution

$$TC = \frac{60000}{12} + \frac{12000}{12} + 2944 = 8944$$

$$Profit = \frac{8944}{44720} = 0.20$$

$$Revenue = (12 \times 6 \times 90) + (5 \times 8 \times 84) + (7 \times 6 \times 84) = 13368$$

$$Profit = 13368 - 8944 - 1200 = 3224$$

Calculation of Ton Kms

	No.	×	Kms	×	Ton	=	Ton Kms
A to B	10	×	300	×	6	=	18,000
A to C	2	×	140	×	6	=	1,680
C to B	2	×	160	×	4	=	1,280
B to A	5	×	300	×	8	=	12,000
B to C	1	×	160	×	6	=	960
C to A	1	×	140	×	0	=	0
B to A	6	×	300	×	6	=	10,800
Total						=	44,720

Statement of operating cost

Particulars	Amount (₹)
Fixed charges (60,000 ÷ 12)	5,000
Maintenance charges (12,000 ÷ 12)	1,000
Running charges	2,944
Total Cost	8,944
Ton-Km	44,720
Total cost per ton-km	0.20

Statement of Profit

Particulars		Amount (₹)
Revenue:		
Outward	$10 \times 6 \times 90 = 5,400$	
	$2 \times 6 \times 90 = 1,080$	
Return	$5 \times 8 \times 84 = 3,360$	
	$1 \times 6 \times 84 = 504$	
	$6 \times 6 \times 84 = 3,024$	13,368
Less: Total cost		(8,944)
Less: Fine & Penalties		(1,200)
Profit		3,224

Question – 7

A transport company has a fleet of three trucks of 10 tonnes, capacity each plying in different directions for transport of customer goods. The trucks run loaded with goods and return empty. The distance travelled, number trips made and the load carried per day by each truck are as under:

Truck No.	One way distance (Km)	No. of trips per day	Load carried per trip per day (tonnes)
1	16	4	6
2	40	2	9
3	30	3	8

The analysis of maintenance cost and the total distance traveled during the last two years is as under:

V. M.C. P. Km = $\frac{46050 - 45175}{160200 - 156700} \text{ Year} = 0.25$	Total distance traveled	Maintenance cost ₹
	1,60,200	46,050
F. M.C. = $46050 - (160200 \times 0.25) = 6000$	1,56,700	45,175
		$(134784 \times 0.25) + 6000 = 39696$

The following are the details of expenses for the last year under review:

Diesel (✓) :	₹ 10 per litre. Each liter gives 4 km per litre of diesel on average	$\frac{10}{4} \times 134784 = 336960$
Driver Salary (F) :	₹ 2,000 per month $\times 12 \times 3 = 72000$	
License and taxes (F) :	₹ 5,000 per annum per truck $\times 3 = 15000$	
Insurance (F) :	₹ 5,000 per annum for all 3 vehicles = 5000	
Purchase price per truck (F) :	₹ 3,00,000, Life 10 years, Scrap value at the end of life is ₹ 10,000	$\frac{(32 - 0.10)}{10} \times 3 = 87000$
Oil and sundries (✓) → :	₹ 25 per 100 km run $\rightarrow \frac{25}{100} \times 134784 = 33696$	
General overhead (F) → :	₹ 11,084 per annum	

The vehicles operate 24 days per month on an average

Required:

- Prepare an Annual Cost Statement covering the fleet of three vehicles
- Calculate the cost per Km run $\rightarrow \frac{600436}{134784} = 4.45$
- Determine the freight rate per tonne km to yield a profit of 10% on freight

$$\frac{(600436)}{525312} \text{ TC / ton-km} = 1.143$$

$$\text{TC} \times \left(\frac{1.143}{90\%} \right) = 1.27$$

Solution**Calculation of Ton Kms**

Truck No.	No.	×	Kms	×	Ton	=	Ton Kms
1	1	×	16 × 4 × 24 × 12	×	6	=	1,10,592
1	1	×	16 × 4 × 24 × 12	×	0	=	0
2	1	×	40 × 2 × 24 × 12	×	9	=	2,07,360
2	1	×	40 × 2 × 24 × 12	×	0	=	0
3	1	×	30 × 3 × 24 × 12	×	8	=	2,07,360
3	1	×	30 × 3 × 24 × 12	×	0	=	0
Total						=	5,25,312

Total kms travelled = **1,34,784 kms****(a) Statement of Operating Cost**

Particulars	Amount (₹)
Fixed Cost:	
Driver salary (2,000 × 12 × 3)	72,000
License and tax (5,000 × 3)	15,000
Insurance	5,000
Depreciation $\left[\frac{(3,00,000 - 10,000)}{10} \times 3 \right]$	87,000
General overheads	11,084
Total Fixed Cost (A)	1,90,084
Variable Cost:	
Diesel [1,34,784 × (10/4)]	3,36,960
Oil & Sundries [1,34,784 × (25/100)]	33,696
Total Variable Cost (B)	3,70,656
Maintenance Cost:	
Maintenance Cost (working note – 1)	39,696
Total Maintenance Cost (B)	39,696
Total Cost (A + B + C)	6,00,436

Working Note 1

$$\text{Variable maintenance cost per km} = \frac{\text{Difference in total cost}}{\text{Difference in kms}} = \frac{46,050 - 45,175}{1,60,200 - 1,56,700} = ₹ 0.25$$

$$\text{Fixed maintenance cost} = \text{Total cost} - \text{Variable cost} = 45,175 - (1,56,700)(0.25) = ₹ 6,000$$

$$\text{Maintenance cost} = \text{Fixed cost} + \text{variable cost} = 6,000 + (1,34,784)(0.25) = ₹ 39,696$$

$$(b) \text{ Total cost per km} = \frac{\text{Total cost}}{\text{Kms}} = \frac{6,00,436}{1,34,784} = ₹ 4.45$$

$$(c) \text{ Cost per ton-km } \left(\frac{6,00,436}{1,34,784} \right) = 1.143$$

$$\text{Profit per ton-km } (1.27 \times 10\%) = 0.127$$

$$\text{Freight per ton-km } (1.143 \div 90\%) = 1.270$$

Question – 8

Due to frequent power cuts and inconsistent water supply from the Electricity Board, Anju Manufacturing Pvt. Ltd. — a mid-sized engineering goods manufacturer based in Pune — decided to set up its own captive power generation plant. The plant is critical for ensuring uninterrupted production across its two major units:

- Unit 1: Produces industrial machinery components (highly power-intensive operations)
- Unit 2: Manufactures agricultural equipment parts (moderate power load)

The factory operations heavily depend on continuous electric and water supply for running CNC machines, foundry operations, welding units, and assembly lines. Any electricity failure leads to major production delays, idle labour costs, and penalties from clients due to late deliveries.

After analysing the risk of depending entirely on the electricity board, the company's Board of Directors approved a proposal to invest in a dedicated steam boiler and power generation plant within the factory premises.

The following information has been taken from the records in connection with the generation of power during the month:

- Coal used 1,800 Tonnes @ ₹230 per tonne → 4,14,000
- Freight and handling charges: 7% of the value of coal used. $\times 4.14 = 28,980$
- Oil: 15 Tonnes @ ₹4,500 per tonne → 67,500
- Water: 80,000 litres @ ₹0.07 per litre → 5,600
- Steam boiler cost ₹50,00,000 with a residual value of ₹2 lakh after a life of 10 years. $\frac{(50-2)}{10 \times 12} = 40,000$
- Salaries and wages of the boiler house:
 - 15 skilled workers @ ₹1,300 per month each → 19,500
 - 30 semi-skilled workers @ 1100 per month each → 33,000
 - 50 unskilled workers @ ₹900 per month each → 45,000
- Recovery on account of sale of ashes: 150 tonnes ₹60 per tonne. → -9,000
- Salary and wages of generating station:
 - 60 skilled workers @ ₹1,200 per month each → 72,000
 - 20 semi-skilled workers @ 1000 per month each → 20,000
 - 30 unskilled workers @ ₹850 per month each → 25,500
- Repairs and maintenance of generating equipment: ₹30,000 →
- Depreciation of generating equipment: ₹75,000 ✓
- Insurance premium on generating plant: ₹15,000 ✓
- Fuel handling and storage expenses: ₹8,000 ✓
- Rent of generating station premises: ₹12,000 ✓
- Share of administration charges: ₹65,000 ✓

- Total number of units generated (before losses): 2,00,000 units
- Normal losses in the process: 3,000 units generated. $\rightarrow 197000 \rightarrow 202000 - 10\% = 181800$
- Electricity purchased from grid to meet emergency requirements: 5,000 units @ ₹6 per unit. $\rightarrow 30000$
- 10% of the effective units generated are used internally by the generator department itself

Additional Information:

- The purchased electricity units are not subject to any loss adjustment.
- Emergency purchase units should be added to the effective units available for use.

You are required to calculate the total cost of electricity generated and the cost of generating electricity per unit. $\rightarrow TC = 997080$
 $\rightarrow \frac{997080}{181800} = 5.48$

Solution

Operating Cost Sheet

Particulars		Amount (₹)
Boiler House Cost		
Coal consumed (1,800 tonnes @ ₹ 230)	4,14,000	
Freight, loading and unloading charges (7% of coal cost)	28,980	
Oil consumed (15 tonnes @ ₹ 4,500)	67,500	
Water consumed (80,000 litres @ Re. 0.07)	5,600	
Depreciation of steam boiler $(\frac{50,00,000 - 2,00,000}{10 \times 12})$	40,000	
Salaries and wages of boiler house workers:		
Skilled workers (15 @ ₹ 1,300)	19,500	
Semi-skilled workers (30 @ 1,100)	33,000	
Unskilled workers (50 @ ₹ 900)	45,000	
Less: Recovery from sale of ash (150 tonnes @ ₹ 60)	(9,000)	
Total Cost (I)		6,44,580
Generating Station Costs		
Salaries and wages of generating station staff:		
Skilled workers (60 @ ₹ 1,200)	72,000	
Semi-skilled workers (20 @ 1,000)	20,000	
Unskilled workers (30 @ ₹ 850)	25,500	
Repairs and maintenance of generating equipment	30,000	
Depreciation of generating equipment	75,000	
Insurance premium on generating plant	15,000	
Fuel handling and storage expenses	8,000	

Rent of generating station premises	12,000	
Share of administration and general expenses	65,000	
Total Cost (II)		3,22,500
Other Costs		
Electricity purchased from grid (5,000 units @ ₹ 6)	30,000	
Total Cost (III)		30,000

Total Cost of Electricity Generated and Purchased (I+II+III)		9,97,080
Units Available (W.N.)		1,81,800 units
Cost per unit		5.48 per unit

Working Note: Units Available

Particulars	Units
Total Units Generated	2,00,000
Less: Normal loss in generation	(3,000)
Net Own Units Generated	1,97,000
Add: Units purchased from grid	5,000
Total Effective Units Available	2,02,000
Less: 10% Units used by generating department	(20,200)
Units Available for Factory Operations	1,81,800

Question – 9

SK Hospital runs a Critical Care Unit (CCU) in a hired building. CCU consists of 35 beds and 5 more beds can be added, if required.

- ~~Rent~~ per month (F) - ₹ 75,000 $\times 12 = 9.0$
~~Supervisors~~ (2 persons) (F) - ₹ 25,000 per month – each $\times 12 \times 2 = 6.0$
~~Nurse~~ (4 persons) (F) - ₹ 20,000 per month – each $\times 12 \times 4 = 9.60$
~~Ward Boys~~ (4 persons) (F) - ₹ 5,000 per month – each $\times 12 \times 4 = 2.40$
 (V) Doctors paid ₹ 2,50,000 per month – paid on the basis of number of patients attended and the time spent by them. $\times 12 = 30.0$

Other expenses for the year are as follows:

- ~~Repair~~ (Fixed) - ₹ 81,000
~~Food~~ to patients (variable) - ₹ 8,80,000
~~Other services~~ to patients (variable) - ₹ 3,00,000
~~Laundry~~ charges (variable) - ₹ 6,00,000
~~Medicines~~ (variable) - ₹ 7,50,000
~~Other fixed~~ expenses (F) - ₹ 10,80,000
~~Administration~~ expenses allocated (F) - ₹ 10,00,000

$$\begin{aligned}
 F &= 4861000 \\
 V &= 5605000 \\
 \hline
 TC &= 10466000
 \end{aligned}$$

$$\begin{aligned}
 \text{Cost. Per Pat. Day} &= \frac{(8000 \times 2000) - 5605000}{8000} \\
 &= 1299.375
 \end{aligned}$$

$$\text{Pat. Days} = (150 \times 35) + (80 \times 25) + 750 = 8000 \checkmark$$

It was estimated that for 150 days in a year 35 beds are occupied and for 80 days only 25 beds are occupied.

The hospital hired 750 beds at a charge of ₹ 100 per bed per day, to accommodate the flow of patients. However, this does not exceed more than 5 extra beds over and above the normal capacity of 35 beds on any day.

You are required to:

(a) Calculate profit per patient day, if the hospital recovers on an average ₹ 2,000 per day from each patient.

(b) Find out breakeven point for the hospital.

$$\frac{(8000 \times 2000) - 10466000}{8000} = \frac{5534000}{8000} = 691.75$$

$$\frac{FC}{\text{Cont. P.V.}} = \frac{4861000}{1299.375} = 3741$$

Solution

(a) Effective bed days = $(150 \times 35) + (80 \times 25) + 750 = 8,000$

Statement of Profit

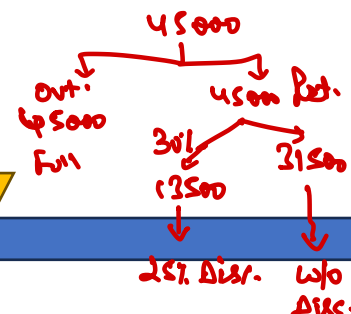
Particulars	Amount (₹)
Variable Cost:	
Doctor cost $(2,50,000 \times 12)$	30,00,000
Food to patients	8,80,000
Other services to patients	3,00,000
Laundry charges	6,00,000
Medicines	7,50,000
Bed Charges (750×100)	75,000
Total Fixed ^{Var.} Cost (A)	56,05,000
Fixed Cost:	
Rent $(75,000 \times 12)$	9,00,000
Supervisor $(25,000 \times 12 \times 2)$	6,00,000
Nurse $(20,000 \times 12 \times 4)$	9,60,000
Ward boys $(5,000 \times 12 \times 4)$	2,40,000
Repair	81,000
Other fixed expenses	10,80,000
Administration expenses	10,00,000
Total Variable ^{Fix.} Cost (B)	48,61,000
Total Cost (A + B)	1,04,66,000
Revenue $(8,000 \times 2,000)$	1,60,00,000
Profit	55,34,000

$$\text{Profit per patient day} = \frac{55,34,000}{8,000} = ₹ 691.75$$

(b) Contribution = Revenue – Variable cost = ₹ 1,60,00,000 – ₹ 56,05,000 = ₹ 1,03,95,000

$$\text{Contribution per patient day} = \frac{1,03,95,000}{8,000} = ₹ 1,299.375$$

$$\text{Break-even point} = \frac{\text{Fixed cost}}{\text{Contribution per patient day}} = \frac{48,61,000}{1,299.375} = 3,741 \text{ patient days}$$



$$23.402 = (y \times 45000) + (45000 \times 30\% \times 0.75y) + (31500 \times y) \Rightarrow y = \frac{23.402}{86625} = 27.01$$

Question – 10

RST Toll Plaza Limited built an 80-kilometer-long highway between two cities and operates a toll plaza to collect tolls from passing vehicles using the highway. The company has estimated that 50,000 light weight, 12,000 medium weight and 10,000 heavy weight vehicles will be using the highway in one month in outward journey and the same number for return journey. - 10% = 45000

As per government notification, vehicles used for medical emergencies, Members of Parliament, and essential services are exempt from toll charges. It is estimated that 10% of light weight vehicles will pass the highway for such use.

It is the policy of the company that if vehicles return within 24 hours of their outward journey, the toll fare will be reduced by 25 percent automatically. It is estimated that 30% of chargeable light weight vehicles return within the specified time frame. for Return fare only

$$\begin{array}{lll} \text{LW} \rightarrow 1 & 90000 & = 90000 \\ \text{MW} \rightarrow 2.5 & 24000 & = 60000 \\ \text{HW} \rightarrow 5 & 20000 & = 100000 \end{array}$$

The toll charges for medium weight vehicles is to be fixed as 2.5 times of the light weight vehicles and that of heavy weight vehicles as 2 times of the medium weight vehicles. ~~2,50,000~~

The toll and maintenance cost for a month is ₹ 59,09,090. The company requires a profit of 10% over the total cost to cover interest and other costs.

- Required:
- $$\text{LW} = \frac{(5909090 + 10\%)}{2.501} = 26 \quad \text{MW} = 26 \times 2.5 = 65 \quad \text{HW} = 26 \times 5 = 130$$
- Calculate the toll rate for each type of vehicle if concession facilities are not available on the return journey.
 - Calculate the toll rate that will be charged from light weight vehicles if a return journey concession facility is available, assuming that the revenue earned from light weight vehicles calculated in option (i) remains the same. $90000 \times 26 = 23.402$

Solution**Working Notes:**

(1) Calculation of equivalent number of light vehicles

Type of vehicles	Monthly traffic (A)	Return traffic (B)	Ratio (C)	Equivalent light weight [(A+B)×C]
Light weight	45,000*	45,000	1	90,000
Medium weight	12,000	12,000	2.5	60,000
Heavy weight	10,000	10,000	5	1,00,000
				<u>2,50,000</u>

*50,000 light vehicles less 10% exempted vehicles.

(2) Calculation of equivalent number of light weight vehicles

Type of vehicles	Monthly traffic (A)	Return traffic (B)	Ratio (C)	Equivalent light weight [(A+B)×C]
Light weight	45,000*	41,625 (45,000 – [45,000 × 30% × 25%])	1	86,625
Medium weight	12,000	12,000	2.5	60,000
Heavy weight	10,000	10,000	5	1,00,000
				2,46,625

$$(i) \text{ Toll rate per vehicle} = \frac{\text{Total cost to cover}}{\text{Equivalent type of vehicles}} = \frac{(59,09,090 + 10\%)}{2,50,000} = \frac{65,00,000}{2,50,000} = ₹ 26$$

$$\text{Toll rate for light weight vehicle} = ₹ 26$$

$$\text{Toll rate for medium weight vehicle} = ₹ 26 \times 2.5 = ₹ 65$$

$$\text{Toll rate for heavy weight vehicle} = ₹ 26 \times 5 = ₹ 130$$

$$(ii) \text{ Revenue from light weight vehicle in (i) above} = 90,000 \text{ vehicles} \times ₹ 26 = ₹ 23,40,000$$

$$\text{New toll rate to maintain the same revenue from Light weight vehicle} = \frac{23,40,000}{86,625} = ₹ 27.01$$

$$\text{Rate to be charged from 13,500 light weight vehicles} = 27.01 \times 0.75 = ₹ 20.26$$

Toll to be charged from light weight vehicles if concession applicable

$$\text{Revenue share in light vehicles} = 90,000 \times 26 = ₹ 23,40,000$$

Suppose rate is x,

then outward journey 45,000 x;

return journey (45,000 – 30% of 45,000) + 13,500(x – 0.25)

$$45,000x + 31,500x + 13,500(0.75x) = 23,40,000$$

$$\text{Toll rate to be charged from light weight vehicles: } 86,625x = 23,40,000$$

$$X = ₹ 27.01$$

$$\text{Rate to be charged from 76,500 light weight vehicles @ ₹ 27.01; revenue will be ₹ 20,66,494}$$

$$\text{Rate to be charged from 13,500 light weight vehicles} = 27.01 \times 0.75 = 20.26$$

$$\text{Revenue will be ₹ 2,73,506}$$

Question – 11

Following are the data pertaining to SK Pvt. Ltd. for the year:

Particulars	Amount (₹)
Salary to Software Engineers (5 persons)	15,00,000
Salary to Project Leader (2 persons)	9,00,000
Salary to Project Manager	6,00,000
Repairs & Maintenance	3,00,000
Administration Overheads	12,00,000

The company executes a Project PK, the details of the same are as follows:

$$\text{OH AP} = \frac{15\%}{30\%} \times 100 = 50\%$$

$$\text{OH} \rightarrow 50\% \times (4.50 + 2.25) + 12 = 13.75$$

Project duration – 6 months

One Project Leader and three Software Engineers were involved for the entire duration of the project, whereas Project Manager spends 2 months' efforts, during the execution of the Project.

Travel expenses incurred for the project – ₹ 1,87,500

Two laptops were purchased at a cost of ₹ 50,000 each, for use in the project and the life of the same is estimated to be 2 years.

$$\frac{50000 \times 2}{2} \times \frac{6}{12} = 25000$$

Prepare project cost sheet.

Solution

Project Cost Sheet

Particulars	Amount
Salary of software engineer ($15,00,000 \times 3/5 \times 6/12$)	— 4,50,000
Salary of project leader ($9,00,000 \times 1/2 \times 6/12$)	— 2,25,000
Salary of project manager ($6,00,000 \times 2/12$)	— 1,00,000
Total salary	7,75,000
Travel expenses	— 1,87,500
Depreciation ($50,000 \times 2 \times 1/2 \times 6/12$)	— 25,000
Overheads ($50\% \times 7,75,000$)	3,87,500
Total Project cost	13,75,000

Working Note:

Total overheads = 3,00,000 + 12,00,000 = ₹ 15,00,000 ✓

Total Salary = 15,00,000 + 9,00,000 + 6,00,000 = ₹ 30,00,000 ✓

Overheads recovery rate = $\frac{\text{Overheads}}{\text{Salary}} \times 100 = \frac{15,00,000}{30,00,000} \times 100 = 50\% \text{ of salary}$

Ratio
② Season. $50 \times 30 \times 6 \times 80\% = 7200$
① Off. Season. $50 \times 30 \times 6 \times 40\% = 3600$
Eq. off season from days
 $= (7200 \times 2) + 3600 = 18000$

Question – 12

A lodging home is being run in a small hill station with 50 single rooms. The home offers concessional rates during six off-season months in a year. During this period, half of the full room rent is charged. The management's profit margin is targeted at 20% of the room rent. The following are the cost estimates and other details for the year ending 31st March, (assume a month to be of 30 days):

(a) Occupancy during the season is 80%, while in the off season is 40% only:

(b) Expenses:

	₹
(1) Staff Salary (excluding room attendants)	2,75,000
(2) Repairs to Buildings	1,30,500
(3) Laundry & Linen	40,000
(4) Interior and Tapestry	87,500
(5) Sundry Expenses	95,400
	628,400

(c) Annual depreciation is to be provided for buildings at 5% and on furniture and equipments at 15% on straight line basis:

$$80 \times 5\% = 40$$

$$20 \times 15\% = 30$$

(d) Room attendants are paid ₹ 5 per room day on the basis of occupancy of the rooms in a month

$$10800 \times 5 = 54000$$

Cost & Management Accounting

Season. $\frac{120}{30} \times 7200 = 28800$
Off. Season. $\frac{120}{30} \times 3600 = 14400$
Total = 43200
Profit = 36000
Net = 24000

- (e) Monthly lighting charges are ₹ 120 per room, except in four months of winter when it is ₹ 30 per room and this cost is on the basis of full occupancy for a month, and
- (f) Total investment in the home is ₹ 100 lakhs of which ₹ 80 lakhs relate to buildings and balance for furniture and equipments.

You are required to compute the room rent per day both during the season and off season.

$$TC = 14,18,400 \quad \text{Rev.} = \frac{14,18,400}{80\%} = 17,73,000$$

Solution

Computation of Effective room days

Season = (50 rooms × 80 / 100) × (6 × 30) days

Off-season = (50 rooms × 40 / 100) × (6 × 30) days

$$\text{R.R. Per Day off. Seas.} = \frac{17,73,000}{18,000} = 98.5$$

$$\text{Seas.} = 98.5 \times 2 = 197$$

Computation of Total Cost

	₹
(1) Salary	2,75,000
(2) Repairs	1,30,500
(3) Laundry and Linen	40,000
(4) Interior Decoration	87,500
(5) Depreciation	
- Building (5% of 80,00,000)	4,00,000
- Furniture & equipment (15% of 20,00,000)	3,00,000
Sundry Expenses	95,400
(6) Attendant's Salary (10,800 × 5)	54,000
(7) Lighting Charges	
- Season (7,200 days × ₹ 4) [₹ 120 p.m. means ₹ 4 per day]	28,800
- Off-Season	
Winter [3,600 × 4/6] × ₹ 1 [₹ 30 p.m. means Re.1 per day] (4 months)	2,400
Balance (3,600 × 2/6) × ₹ 4 (2 months)	4,800
Total Cost	14,18,400

Computation of Total Revenue

	₹
Total Cost	14,18,400
(+) Profit (20% of revenue) (14,18,400 × 20/80) -	3,54,600
Total Revenue	17,73,000

Assume Rent per room per day during Season is ₹ X & during off season is ₹ X/2

Hence, total annual revenue = 7,200 X + 3,600 [X/2] = 9,000 X

Now, 9,000 X = ₹ 17,73,000

X = 197 Hence, Rent per room per day

During Season = X = ₹ 197

During off-season = [X/2] = ₹ [197/2] = ₹ 98.50

Question – 13

ABC Bank is having a branch which is engaged in processing of 'Vehicle Loan' and 'Education Loan' applications in addition to other services to customers. 30% of the overhead costs for the branch are estimated to be applicable to the processing of 'Vehicle Loan' applications and 'Education Loan' applications each.

Branch is having four employees at a monthly salary of ₹ 50,000 each, exclusively for processing of Vehicle Loan applications and two employees at a monthly salary of ₹ 70,000 each, exclusively for processing of Education Loan applications.

In addition to above, following expenses are incurred by the Branch:

- Branch Manager who supervises all the activities of branch, is paid at ₹ 90,000 per month.
- Legal charges, Printing & stationery and advertising expenses are incurred at ₹ 30,000, ₹ 12,000 and ₹ 18,000 respectively for a month.
- Other expenses are ₹ 10,000 per month.

You are required to:

- Compute the cost of processing a Vehicle Loan application on the assumption that 496 Vehicle Loan applications are processed each month.
- Find out the number of Education Loan applications if the total processing cost per Education Loan Application is same as in the Vehicle loan Application as computed in (i) above.

$$500 = \frac{1,880}{\text{No. of App.}} \Rightarrow \text{No. of App.} = \frac{1,880}{500} = 376$$

Solution

Particulars	Vehicle Loan Applications (₹)	Education Loan Applications (₹)	Total (₹)
Employee Cost	50,000 × 4 = 2,00,000	70,000 × 2 = 1,40,000	3,40,000
Apportionment of branch manager's salary	27,000	27,000	54,000
Legal charges, printing & stationary and advertising	18,000	18,000	36,000
Other expenses	3,000	3,000	6,000
Total cost	2,48,000	1,88,000	4,36,000

$$(a) \text{ Cost of processing vehicle loan application} = \frac{\text{Total cost}}{\text{No. of applications}} = \frac{2,48,000}{496} = ₹ 500$$

$$(b) \text{ Cost of processing education loan application} = \frac{\text{Total cost}}{\text{No. of applications}}$$

$$500 = \frac{1,88,000}{\text{No. of applications}}$$

$$\text{No. of applications} = \frac{1,88,000}{500} = 376$$

Question – 14

MRS L Healthcare Ltd. has incurred the following expenditure during the last year for its newly launched 'COVI-19' Insurance policy:

	₹
Office administration cost (S)	48,00,000
Claim management cost (O)	3,80,000
Employees' cost (S)	16,20,000
Postage and logistics (S)	32,40,000
Policy issuance cost (O)	29,50,000
Facilities cost (S)	46,75,000
Cost of marketing of the policy (M)	1,38,90,000
Policy development cost (M)	35,00,000
Policy servicing cost (O)	96,45,000
Sales support expenses (M)	32,00,000
IT cost → (IT)	2,21,00,000 ?

$$TC = IT + \text{Other}$$

$$700\text{L} = IT + 479\text{L}$$

$$IT = 221\text{L}$$

479L

Number of Policy sold: 2,800

Total insured value of policies - ₹3,500 crores

Cost per rupee of insured value - ₹0.002 → $TC = 7\text{L}$

You are required to:

- Calculate the total cost for "COVID-19" Insurance policy segregating the costs into four main activities namely (a) Marketing and Sales support (b) operations (c) IT Cost and (d) Support functions.
- Calculate cost per policy → $\frac{700\text{L}}{2800} = \frac{1}{2} = 25000$

Solution

- Total Cost = Total insured value × Cost per rupee of insured value

$$\text{Total Cost} = ₹3,500 \text{ crore} \times 0.002$$

$$\text{Total Cost} = ₹7,00,00,000$$

$$\text{Other Cost} + \text{IT Cost} = 7,00,00,000$$

$$4,79,00,000 + \text{IT Cost} = 7,00,00,000$$

$$\text{IT Cost} = ₹2,21,00,000$$

Statement of Cost

Particulars	Amount
Marketing and Sales Support:	
Cost of marketing the policy	1,38,90,000
Policy development cost	35,00,000
Sales support expenses	32,00,000
Total (A)	2,05,90,000
Operations Cost:	
Claim management cost	3,80,000
Policy issuance cost	29,50,000
Policy servicing cost	96,45,000
Total (B)	1,29,75,000

IT Cost:	
IT Cost	2,21,00,000
Total (C)	2,21,00,000
Support Function:	
Office administration cost	48,00,000
Employees' cost	16,20,000
Postage and logistics	32,40,000
Facilities cost	46,75,000
Total (D)	1,43,35,000
Total Cost (A + B + C + D)	7,00,00,000
Number of Policies	2,800
Cost per policy	25,000